



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Division: A2

ASSIGNMENT No: 14

TITLE: Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement: Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives: To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory: Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Exercise:

1. Create an ATM program handling invalid withdrawal amount using try-catch

Program:

```
C:\Users\arhaa\OneDrive\Desktop\java\main.java
2
3  class ATM {
4      public static void main(String[] args) {
5          int balance = 10000;
6          try (Scanner sc = new Scanner(System.in)) {
7              System.out.print(s: "Enter withdrawal amount: ");
8              int amount = sc.nextInt();
9              if (amount <= 0 || amount > balance) {
10                 throw new Exception(message: "Invalid withdrawal amount!");
11             }
12             balance -= amount;
13             System.out.println(x: "Withdrawal Successful.");
14             System.out.println("Remaining Balance: " + balance);
15         } catch (Exception e) {
16             System.out.println(e.getMessage());
17         }
18     }
19 }
```

Output:

```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\arhaa\OneDrive\Desktop\java> &
ng\Code\User\workspaceStorage\02e7f2cfce92a
Enter withdrawal amount: 1500
Withdrawal Successful.
Remaining Balance: 8500
PS C:\Users\arhaa\OneDrive\Desktop\java>
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

Program:

```
1 C:\Users\arhaa\OneDrive\Desktop\java\main.java
2
3 class Shopping {
4     Run | Debug
5     public static void main(String[] args) {
6         try (Scanner sc = new Scanner(System.in)) {
7             System.out.print(s: "Enter product quantity: ");
8             int quantity = sc.nextInt();
9             if (quantity <= 0) {
10                 throw new Exception(message: "Quantity must be greater than zero.");
11             }
12             System.out.println(x: "Order placed successfully.");
13         } catch (Exception e) {
14             System.out.println(e.getMessage());
15         }
16     }
17 }
```

Output:

```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Code\User\workspaceStorage\02e7f2cfce92a0ba8e306
Enter product quantity: 25
Order placed successfully.
PS C:\Users\arhaa\OneDrive\Desktop\java> █
```